

A Literature on Deficit Financing and Fiscal Stability

George Chou

School of Public Finance and Taxation, ZUEL

April 2nd, 2026



Outline

- 1 The constraint on public debt when $r < g$ but $g < m$
- 2 Can deficits finance themselves?
- 3 A goldilocks theory of fiscal deficits
- 4 Sovereign risk, fiscal policy, and macroeconomic stability
- 5 The liquidity channel of fiscal policy

Ricardian Equivalence

- Does it matter whether the government finances its expenditures with debt or (nondistortionary) lump-sum taxes?
- Barro (1974) showed that the means of financing public expenditures does not matter for the real allocation of the economy if the following holds:
 - ① Families act as infinitely lived dynasties because of intergenerational altruism.
 - ② Capital markets are perfect (i.e., all can borrow and lend at a single rate). In particular, private and public debt are perfect substitutes.
 - ③ The path of government expenditures is fixed. Therefore, the government can precommit itself and its successors to a specific fiscal policy.

Ricardian Equivalence

- Let consumption paths $\{c_t\}_{t=0}^{\infty}$ be elements of a Banach space $\mathcal{X}$ (e.g., l^p).
- Private sector wealth is defined by the inner product $\langle \mathbf{p}, \mathbf{c} \rangle = W$, where $\mathbf{p}$ is the pricing kernel.
- **Barro-Ricardo Equivalence Theorem:** Ricardian Equivalence holds if the government's intertemporal budget constraint (GIBC) acts as a **translation operator** that leaves the consumer's feasible set $\mathcal{F}$ invariant.

Functional Representation

The timing of taxes τ vs. debt d is irrelevant if:

$$\langle \mathbf{p}, \mathbf{c} \rangle \leq \langle \mathbf{p}, \mathbf{y} - \mathbf{g} \rangle$$

where the sequence of government spending $\mathbf{g}$ is fixed, rendering the choice between τ and d a mere redistribution within the same subspace.

Failure of Ricardian Equivalence

- Finite Horizon;
- Distortionary Taxation;
- Incomplete Markets;
- Borrowing Constraint;
- Failure of Ramsey Plan.

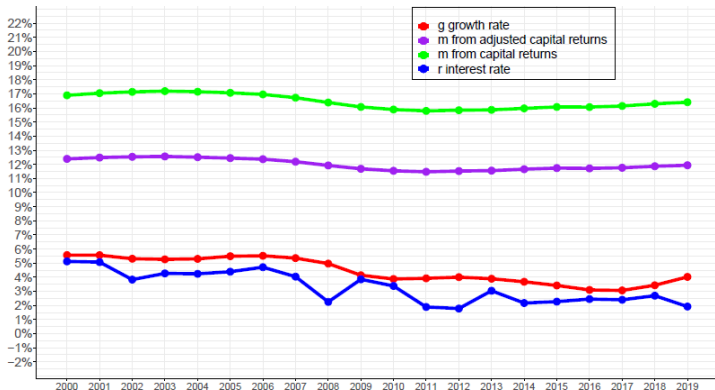
Equivalence of Several Models

- Aggrawal et al. (2023) had shown that, these models below share the same iMPC matrix, that is, to first order they have identical responses to income shocks at any date.
 - Bond-in-utility;
 - Blanchard prepetual-youth OLG;
 - Rotemberg capital adjustment cost.

- 1 The constraint on public debt when $r < g$ but $g < m$
- 2 Can deficits finance themselves?
- 3 A goldilocks theory of fiscal deficits
- 4 Sovereign risk, fiscal policy, and macroeconomic stability
- 5 The liquidity channel of fiscal policy

- **Title:** The constraint on public debt when $r < g$ but $g < m$
- **Publication:** CEPR Discussion Paper, 2021
- **Author:** Ricardo Reis, LSE

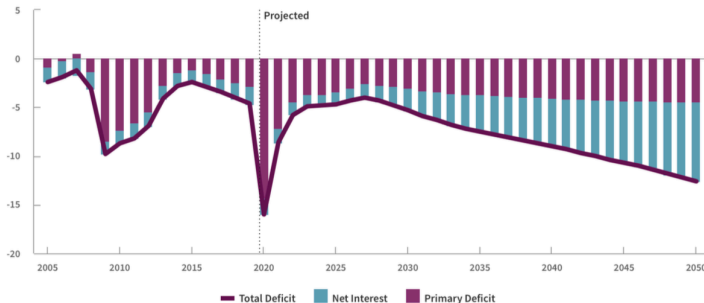
Stylized Fact



Stylized Fact

Total Deficits, Primary Deficits, and Net Interest

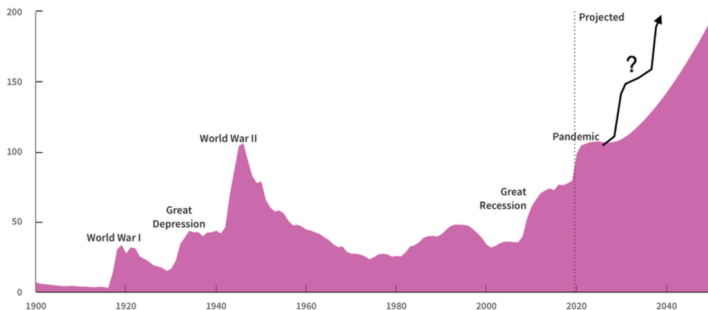
Percentage of Gross Domestic Product



Stylized Fact

Federal Debt Held by the Public, 1900 to 2050

Percentage of Gross Domestic Product



Deficit Stability

- The government can run a perpetual budget deficit when $r < g < m$, but there is still a well-defined budget constraint whereby the debt equals the present value of the ratio of primary surpluses to output discounted by $m - g$ plus the discounted bubble premia earned on the debt that equals $m - r$.
- It is not the gap $r - g$, but rather the gaps $m - g$ and $m - r$ that matter for public finances.
- Questions:
 - why is $m > g > r$ and so what drives the two gaps?
 - How does more government spending affect the bubble premium in equilibrium?
 - Is there an upper bound on the amount of spending for the bubble to be sustainable?
 - How do monetary and fiscal policies affect the bubble premium, and through it do they tighten or loosen the government budget constraint?

- The law of motion for the evolution of the public debt:

$$db_t = s_t dt + r_t b_t dt. \quad (1)$$

- Assumptions:
 - The primary deficit s_t is exogenous.
 - There is no aggregate uncertainty affect either r_t or s_t .
 - Output y_t grows at the rate g_t .
- Empirical evidence shows that $r_t \leq g_t$ and $b_t \geq 0$. We also consider the BGP, i.e., exogenous spending s_t and the endogenous debt asymptotically grow at the same rate as output, g .

Permanent Deficits?

- In the BGP, equation (1) implies that

$$b = \max \left\{ \frac{s}{g - r}, 0 \right\}. \quad (2)$$

- In spite of a fixed permanent deficit, with $r < g$, the government can still sustain positive debt.
- Intuition: When $g = 0$, since $r < 0$, the debt holder is paying the government a fixed stream to hold the bond. This revenue is what finances the permanent spending.

Recurrent Deficits?

- Solving the debt dynamics in equation (1) forward to infinity:

$$b_0 = \lim_{T \rightarrow \infty} \left[- \int_0^T e^{-\bar{r}_t t} s_t dt + \int_0^T e^{-\bar{r}_t T} b_T \right].$$

- Since asymptotically $r < g$, $b_0 = +\infty$. That is, any initial debt b_0 can be sustained, with no limit on public debt beyond the available resources in the economy, since deficit gambles can be repeated and rolled over in a Ponzi way.
- INCORRECT! $b_0 = -\infty + \infty = ?$

Recurrent Deficits?

- Rewriting equation (1):

$$db_t - d_t b_t dt = s_t dt + (r_t - d_t) b_t dt,$$

for some discount rate $d_t < g_t, \forall t$. Then the limit are well-defined and

$$b_0 = - \int_0^\infty e^{-\bar{d}_t t} s_t dt + \int_0^\infty e^{-\bar{d}_t t} (d_t - r_t) b_t dt. \quad (3)$$

- Which is an appropriate d_t to use?

- Let m_t be the marginal product of capital in the economy. This gives the private return of investing in production as opposed to in the government debt. Since, in the data $m_t > g_t$, this is an empirically legitimate choice for d_t .
- Equation (3) can be written as

$$b_0 = - \int_0^{\infty} e^{-\bar{m}_t t} s_t dt + \int_0^{\infty} e^{-\bar{m}_t t} (m_t - r_t) b_t dt. \quad (4)$$

- The first term is the present value of spending, using the return on private assets as the valid SDF, as one would for the payoffs in any other asset.
- The second term is the present value of the implicit government revenues that arise from paying r_t in its debt below the marginal return in the private economy m_t .

- The spread between m_t and r_t measures how special debt is: its bubble premium, or convenience yield, or seigniorage from issuing bonds that provide a service.
- When bubble revenue $b_t(m_t - r_t) > 0$, the government can now pay for outstanding debt in part through these bubble revenues, so recurrent spending can be positive in present value. This leads to a free-lunch problem.
- Why there is a spread?

- With incomplete financial markets, the most productive entrepreneurs cannot borrow as much as they would like to invest in their technology. Least productive and risky firms are in business, creating a misallocation of resources that endogenously drives the wedge between the marginal product of capital and the growth rate of the economy.
- Public debt provides:
 - **Safety:** Households are unable to trade the idiosyncratic depreciation risk that they bear if they invest in capital. The public debt is safe, since its returns are uncorrelated with the returns on individual capital.
 - **Liquidity:** It is an alternative to store wealth over periods for households if the constraint on private lending is too tight. Households are aware that they can become high-types in the future, and want to store value for when this happens.

- Two complementary ways for investigation:
 - Keeping spending s/b fixed, asking whether the policy lowers or raises the fiscal capacity $\mathcal{S}$.
 - Let government spending respond to the policy change in order to keep the interest rate fixed, and ask whether the policy leads permanent spending s/b to rise or fall. (s/b means the fiscal space)

Monetary-Fiscal Policy Trade-Offs

- Monetary policy: inflating the debt or deflating the bubble?
 - Changes in expected inflation π_t have no consequence on government spending. A higher variance of inflation σ_π lowers the fiscal capacity and the fiscal space.
- Financial repression: sacrificing private credit.
 - An increase in the share of public debt that is coerced raises the fiscal capacity and the fiscal space.
- Redistributive policy: lowering inequality versus raising public spending.
 - A larger redistribution program lowers the fiscal capacity and the fiscal space.
- Fiscal policy: tax cuts that pay for themselves?
 - A marginal increase in the tax rate on all income that is fully rebated to each type of household, raises fiscal capacity but lowers fiscal space.

New Empirical Challenges

- How much does the bubble premium respond to shocks to government spending?
- How much does it change with other policies?
- How do different compositions of the public debt affect the overall bubble premium?
- Can the three-way interaction between the SDF, the bubble premium, and the amount of debt make quantitative progress on solving the debt valuation puzzle?
- Do the considerations of $r < g < m$ quantitatively change the strategic incentive for countries to default well before they reach their fiscal capacity?

- 1 The constraint on public debt when $r < g$ but $g < m$
- 2 Can deficits finance themselves?
- 3 A goldilocks theory of fiscal deficits
- 4 Sovereign risk, fiscal policy, and macroeconomic stability
- 5 The liquidity channel of fiscal policy

- **Title:** Can deficits finance themselves?
- **Publication:** Econometrica, 2024
- **Authors:**
 - George-Marios Angeletos, Northwestern U and NBER
 - Chen Lian, UC Berkeley and NBER
 - Christian K. Wolf, MIT and NBER

Self-Financing Problem

- Suppose the government runs a deficit today in order to stimulate aggregate demand. Suppose further that there is no free lunch ($r < g$). That is, the government's net cost of borrowing is positive.
- What does the government have to do in order to make sure that public debt eventually returns back to its initial level?
 - Conventional view: Fiscal adjustment. Adopting tax hikes and/or spending cuts in order to pay down the accumulated debt.
 - Alternative view: Self-financing. Insofar as a deficit triggers a boom, it can contribute to its own financing via two complementary channels:
 - ① By expanding the tax base, which helps generate additional tax revenue without any adjustment in tax rates;
 - ② By triggering inflation, which helps reduce the real value of the government's nominal liabilities, i.e., debt erosion.

Self-Financing Problem

- Key features in the model:
 - A failure of Ricardian equivalence. So that deficits can stimulate aggregate demand;
 - Nominal rigidity. So that aggregate demand can drive real activity and thereby inflation.
- The degree of self-financing is positive and increases as the fiscal adjustment is delayed.
- If the monetary authority does not lean too heavily against the fiscal stimulus, and as the fiscal adjustment is delayed more and more, the tax hike needed to bring debt back to trend vanishes.

- Firms: Exactly the same as textbook New Keynesian model, such as Woodford (2003) or Gali (2008). It boils down to the usual NKPC.
- Households: OLG of perpetual-youth a la Blanchard (1985) with survival probability $\omega \in (0, 1]$.
- Government: Determining the rule for how tax revenue (and thereby the primary surplus) responds to changes in aggregate income and in public debt.
- Monetary Authority: The cyclicalities of (expected) real interest rates. Especially, keeping neutral in the baseline model.

Self-Financing Condition

- Starting from steady state, we shock the economy with a one-off, deficit-financed lump-sum transfer to households.
- How much of this deficit shock will be self-financed in equilibrium via the tax base and inflation channels?
- Or, conversely, how large are the tax hikes needed, sooner or later, to bring public debt back to its trend?

Self-Financing Condition

- When $\omega = 1$, Ricardian equivalence holds, and thus aggregate demand is invariant to the initial deficit shock and the timing of the subsequent fiscal adjustment.
- When $\omega < 1$,
 - Ricardian equivalence fails, and the deficit—which now represents a net transfer from future generations to current generations or, less literally, helps some consumers overcome liquidity constraints—stimulates aggregate demand.
 - With nominal rigidities, this increase in aggregate demand translates to an increase in real income and, via the NKPC, to an increase in the nominal price level.
- How large can such self-financing be?

Self-Financing Condition

- Let ν denote the overall degree of self-financing; that is, the fraction of the initial deficit that is financed via the tax base and inflation channels.
- If $\omega < 1$, then ν converges monotonically toward 1 as fiscal adjustment is pushed further into the future.
- Furthermore, the stickier the prices, the larger the share of self-financing through tax base expansion, and vice versa.

Toy Model of Self-Financing

- Consider an analogy with a simple two-period economy, where $t \in \{0, 1\}$.
 - $t = 0$ represents the “short run”, when the effects of the fiscal stimulus play out.
 - $t = 1$ represents the “long run”, or the steady state.
- The government pays out transfers of size $\varepsilon > 0$ at $t = 0$, and hikes taxes as much as necessary to bring total government debt down to its initial level at $t = 1$ (which is unanticipated for consumers).
- Aggregate consumer demand at $t = 0$ is proportional to concurrent disposable income.
- Prices at $t = 0$ are rigid, so output is fully demand-determined.
- The lump-sum tax rate is $\tau_y \in (0, 1)$, where y denote aggregate income at $t = 0$ ¹.

¹Normalized so that $y = 0$ when $\varepsilon = 0$.

Toy Model of Self-Financing

- By applying simple Keynesian cross logic, we infer that

$$y = \frac{\text{MPC}}{1 - \text{MPC}(1 - \tau_y)} \varepsilon.$$

- The output boom generates additional tax revenue of $\tau_y y$, so the degree of self-financing is given by

$$\nu \equiv \frac{\tau_y y}{\varepsilon} = \frac{\tau_y \times \text{MPC}}{1 - \text{MPC}(1 - \tau_y)}. \quad (5)$$

Toy Model of Self-Financing

- Equation (5) shows two properties:
 - ① $\frac{\partial \nu}{\partial \text{MPC}} > 0$. A larger MPC maps to both
 - Larger numerator, PE effect of the transfer on consumer spending.
 - Smaller denominator, stronger GE amplification.
 - ② $\nu \rightarrow 1 \Leftrightarrow \text{MPC} \rightarrow 1$. Every dollar of initial transfer ends up generating exactly one dollar of additional tax revenue, that is, full self-financing obtains.

Intuition in General Environment

- **Discounting:** Consumers discount any future income, and hence also any future taxes, at a higher rate than the interest rate faced by the government. This implies that tax hikes in the far-ahead future have little effect on consumer demand in the short run.
- **Front-Loading:** consumers spend both the initial fiscal transfer as well as any additional income generated in general equilibrium relatively quickly. This implies that, when fiscal adjustment is sufficiently delayed, the cumulative MPC over the short run approaches 1.
- With an upward-sloping aggregate Phillips curve, the stimulus-led boom is now accompanied by a surge in inflation and an erosion in the real debt burden. But this channel does not alter the monotonicity and limit properties of ν .

- 1 The constraint on public debt when $r < g$ but $g < m$
- 2 Can deficits finance themselves?
- 3 A goldilocks theory of fiscal deficits
- 4 Sovereign risk, fiscal policy, and macroeconomic stability
- 5 The liquidity channel of fiscal policy

- **Title:** A goldilocks theory of fiscal deficits
- **Publication:** American Economic Review, 2025
- **Authors:**
 - Atif Mian, Princeton U and NBER
 - Ludwig Straub, Harvard U and NBER
 - Amir Sufi, U Chicago Booth and NBER

- An economy suffers from persistent shortfalls in aggregate demand, causing:
 - Actual growth to remain below potential;
 - Sluggish economic expansion;
 - Subdued inflation;
 - A long-term equilibrium real interest rate (the neutral rate consistent with full employment and stable inflation) that is extremely low or negative.

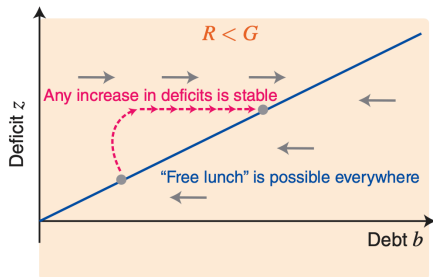
- A prolonged decline in the equilibrium real interest rate r leads to the ZLB, which allows for the possibility that weak demand reduces output and inflation and thus also the nominal growth rate g of the economy.
- The existence of free-lunch: $r < g$, and r increases in government debt since it provides convenience benefits. (bubble premium exists)

Free-Lunch Condition

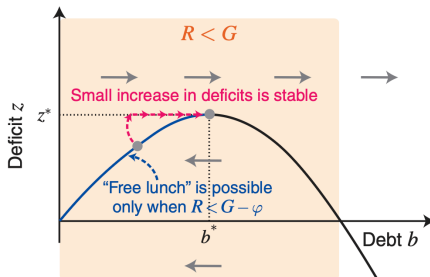
- Echoing Reis (2021), the correct condition for the existence of a free lunch policy is not $r < g$; instead, it is a tighter condition, $r < g - \varphi$, where φ is the sensitivity of $r - g$ to the logarithm of public debt to GDP.
- As a consequence, even for countries in which $r < g$, borrowing more may not be free.
- Suppose that $r < g$. The government decides to borrow 1 additional percent of GDP and plans to roll it over forever.
 - The rolling over of the additional debt produces a positive cash flow for the government equal to $g - r$.
 - The additional debt also tightens the budget constraint because of its impact on the interest rate on all inframarginal outstanding units of debt, which is precisely captured by φ .

Free-Lunch Condition

Panel A. Model with exogenous $R < G$



Panel B. Model with endogenous $R^*(b)$

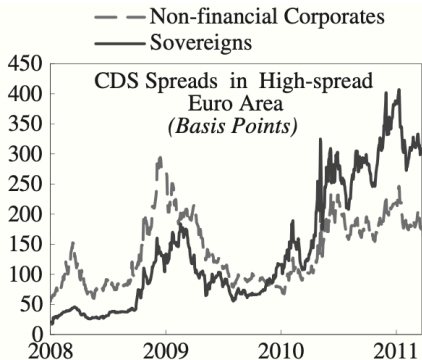
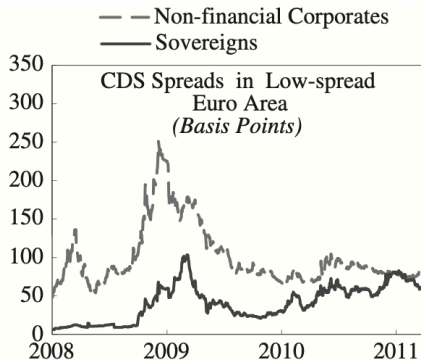


- Opposed to the textbook view, also echoing Angeletos et al. (2024), this paper shows that greater deficits may reduce, rather than increase, debt. (measuring by κ , the slope of NKPC)
 - 1 Greater deficits raise aggregate demand and inflation;
 - 2 Higher inflation translates into higher nominal growth rates;
 - 3 This pushes debt down, as it increases the speed at which debt is “inflated away.”

- 1 The constraint on public debt when $r < g$ but $g < m$
- 2 Can deficits finance themselves?
- 3 A goldilocks theory of fiscal deficits
- 4 Sovereign risk, fiscal policy, and macroeconomic stability
- 5 The liquidity channel of fiscal policy

- **Title:** Sovereign risk, fiscal policy, and macroeconomic stability
- **Publication:** Economic Journal, 2013
- **Authors:**
 - Giancarlo Corsetti, European University Institute and CEPR
 - Keith Kuester, U Bonn and CEPR
 - Andre Meier, Tudor Capital Europe
 - Gernot J. Muller, U Tuebingen and CEPR

Stylized Fact



- In the wake of the global financial crisis, sovereign risk premia have risen sharply in several countries.
- This trend has been accompanied by a marked tightening of private credit markets in the same countries.
- Is there any link between them?
 - Higher public indebtedness adversely affects private-sector financing costs and explore its implications for macroeconomic stability and fiscal stabilization policies.

- Based on Curdia and Woodford (2009), in which heterogeneous households engage in borrowing and lending via financial intermediaries.
- Sovereign risk premia that respond to changes in the fiscal outlook of the country is allowed.
- Private credit spreads rise with sovereign risk because strained public finances raise the cost of financial intermediation.
- Monetary policy would be similarly constrained under a currency peg or in other situations where political or institutional considerations prevent the central bank from counteracting a rise in sovereign risk premia. (e.g. ZLB)

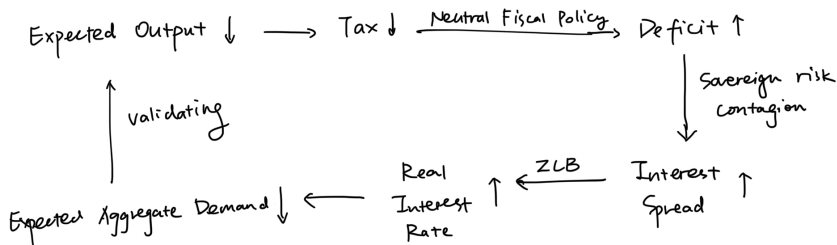
Self-Filling of Expectations

- Under these circumstances, sovereign risk may give rise to indeterminacy, or belief-driven equilibria.
 - Pessimistic shift in expectations → upward revision of the projected government deficit → Increase of private borrowing costs → slow down economic activity → validating the initial adverse shift in expectations.
- How to avoid?
 - Appropriately lower the policy rate through the central bank's commitment.
 - Fiscal tightening in response to a cyclical fall in tax revenue can help to ensure determinacy. (when contagion is strong and monetary policy is strictly constrained)

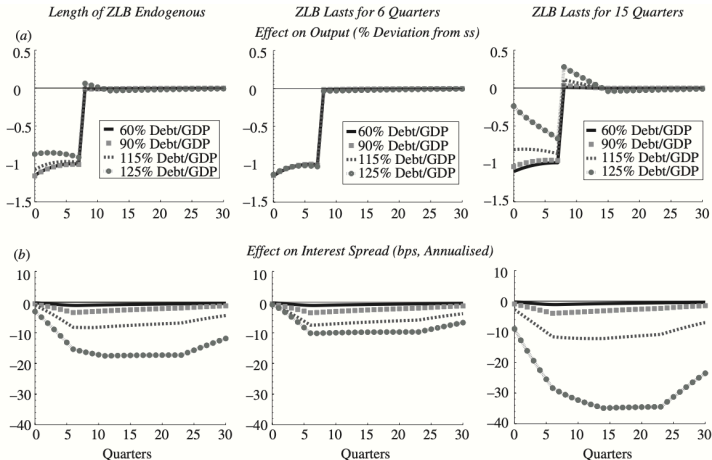
Multiplier Effect of Spending

- The sign and the size of the government spending multiplier depend critically on the state of the economy.
 - When the central bank is unconstrained, the sovereign risk channel is not operative in our model, as looser monetary policy can fully offset the impact of higher risk premia.
 - When the central bank is constrained, the sovereign risk channel tends to reduce the fiscal multiplier.
 - For extreme cases, the multiplier even changes its sign, especially when public finances are very fragile and monetary policy is constrained for an extended period. (pro-cyclical fiscal retrenchment)

Self-Filling of Negative Belief



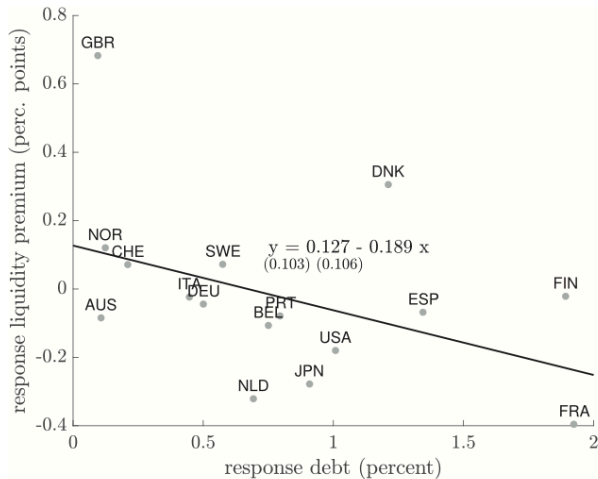
Output and Contagion Effects during ZLB



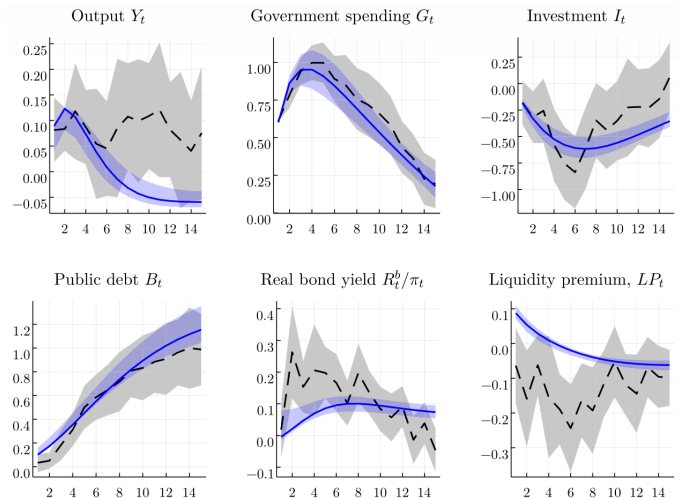
- 1 The constraint on public debt when $r < g$ but $g < m$
- 2 Can deficits finance themselves?
- 3 A goldilocks theory of fiscal deficits
- 4 Sovereign risk, fiscal policy, and macroeconomic stability
- 5 The liquidity channel of fiscal policy

- **Title:** The liquidity channel of fiscal policy
- **Publication:** Journal of Monetary Economics, 2023
- **Authors:**
 - Christian Bayer, U Bonn, CEPR, and IZA
 - Benjamin Born, Frankfurt School of Finance & Management and CEPR
 - Ralph Luetticke, U Tuebingen, CEPR, and CFM

Stylized Fact

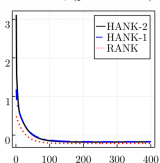


Short-Run IRF

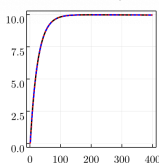


Long-Run IRF

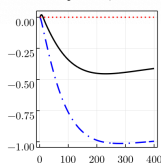
Transfer τ_t (percent of $\bar{Y}$)



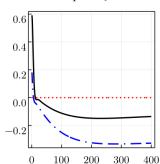
Public debt B_t



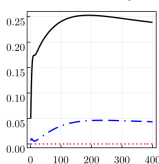
Capital K_t



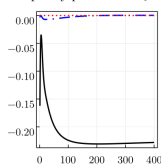
Output Y_t



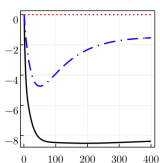
Nominal rate RB_t



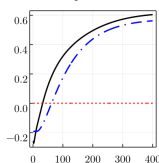
Liquidity premium LP_t



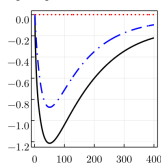
Private debt IOU_t



Consumption Gini



Top 10 percent wealth share



Questions are welcome.